

Dynamics and Reconstruction from Observation: Predictive Operators, the Koopman Picture, and the Density Bridge

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Abstract

Starting from a σ -additive probability measure on an observable algebra — the object constructed in [4] from coherent structured observations satisfying collective exhaustion — this paper derives two structures that are often assumed: the dynamics of prediction, and the recoverability of the state space.

Making prediction precise forces the introduction of the predictive kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$, the regular conditional distribution of a future observation F given a current observation Q . Composing with Q gives the minimal predictive state map $Q_* : \Omega \rightarrow \mathcal{P}(\beta)$. Indexing over time, the kernels compose via Chapman–Kolmogorov — derived, not assumed — yielding a Markov semigroup $\{K_t\}$ and the Koopman–Perron duality $\int K_t g d\mu = \int g d(\mu \star \Pi_t)$. When the kernels are Dirac measures, K_t collapses to the classical Koopman operator.

The construction produces a predictive state space $\mathcal{P}(\beta)$. In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$, the map Q_* becomes the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$. This raises the reconstruction question: when does Φ_h faithfully represent the full state space X ?

The answer is a three-way equivalence — the reconstruction theorem. State-space recovery ($\mathcal{O}_h = \mathcal{B} \bmod \mu$) holds if and only if the delay algebra $\mathcal{A}_h$ is L^2 -dense, if and only if Φ_h is a measure-theoretic embedding. The bridge between the first two conditions is a general lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty$ containing 1, the L^2 -closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. When reconstruction holds, the compact Stone space of [4] — introduced there as a technical device for σ -additivity — is identified with X itself.

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1 Introduction

The classical approach to dynamics, following Koopman [3], describes how observable functions evolve under a known transformation T : the operator $U_T f = f \circ T$ acts by precomposing with T as a primitive. This paper asks two questions that precede this picture.

The first is: given a current observation $Q : \Omega \rightarrow \alpha$ and a future observation $F : \Omega \rightarrow \beta$, what does an observer predict about F given Q ? The dynamics, if there are any, should follow from the answer — not precede it. Making prediction precise immediately requires a σ -additive probability measure on the observable algebra. That measure is not an extra assumption. It is what [4] constructs: a coherent family of finitely additive charges on a directed system of observable algebras extends to a genuine probability measure exactly when it satisfies collective exhaustion, the condition ruling out persistent mass on globally vanishing events.

Once the measure exists, the canonical answer to the prediction question is the regular conditional distribution of F given Q : the unique Markov kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ realising $P(F \in$

$\cdot \mid Q = q$) simultaneously across all outcomes. Its existence rests on the Rokhlin disintegration theorem [5], which requires σ -additivity. Composing with Q gives the minimal predictive state map $Q_* : \Omega \rightarrow \mathcal{P}(\beta)$, the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. From it emerge a semigroup of Markov operators $\{K_t\}$ and the Koopman–Perron duality connecting them with the pushforward operators. When the kernels are Dirac measures, the construction collapses to the classical Koopman picture.

The construction produces a predictive state space $\mathcal{P}(\beta)$ and a stationary distribution ν_{Q_*} . In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$ for some $h \in L^\infty$, the map Q_* specialises to the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$. This raises the second question: when does Φ_h faithfully represent the full state space X ?

The second question is the reconstruction problem. Whether Φ_h is injective modulo null sets — whether the predictive state space faithfully represents X — depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X . The reconstruction theorem gives a three-way equivalence: the σ -algebraic condition $\mathcal{O}_h = \mathcal{B}$ modulo μ , the analytic condition that the delay algebra $\mathcal{A}_h$ is L^2 -dense, and the geometric condition that Φ_h is a measure-theoretic embedding are all equivalent.

The key to the equivalence is the density bridge lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty(X, \mu)$ containing the constant 1, the L^2 -closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. The proof goes via the functional monotone class theorem [1]. The σ -algebra and the L^2 space it generates are inextricably linked.

The Koopman operator $U_T f = f \circ T$ provides a spectral perspective on the second question. The cyclic vector condition — $\overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2$ — implies reconstruction but is strictly stronger: it asks for density of the linear span, while reconstruction asks only for density of the algebra the orbit generates. The two coincide precisely when U_T has simple L^2 -spectrum.

When reconstruction holds, a compact space introduced in [4] for an entirely different purpose — the Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra, built as a technical completion of the sample space for deriving σ -additivity — turns out to be measure-theoretically isomorphic to X itself. The programme that began by seeking a compact ambient space for the measure ends by identifying it with the state space being observed.

The paper is self-contained modulo the probability measure. Sections 2 and 3 develop the predictive structure from a probability space $(\Omega, \mathcal{F}, P)$. Section 4 connects the two parts: the observable dynamical system produced by prediction is the natural input to the reconstruction problem. Sections 5 and 6 carry out the reconstruction, proving the density bridge and the reconstruction theorem.

2 Prediction and the minimal predictive state

Fix a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [5] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ satisfying

$$\int_A \Pi_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 2.1 (Predictive kernel). The *predictive kernel* is Π_Q , the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\Pi_Q(q, A) = P(F \in A \mid Q = q)$.

The kernel is covariant under measurable maps: if $\pi : \alpha \rightarrow \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$ -almost every q , by uniqueness of disintegration.

Composing Π_Q with Q gives the map sending each sample point to its conditional law.

Definition 2.2 (Minimal predictive state map). The *minimal predictive state map* is

$$Q_* : \Omega \rightarrow \mathcal{P}(\beta), \quad Q_*(\omega) = \Pi_Q(Q(\omega), \cdot).$$

Since Π_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 2.3 (Predictive factorization). *For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] = \int g dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g d\Pi_Q(q, \cdot)$ is strongly measurable; since $Q_* = \Pi_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$ -measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned} \int_S h dP &= \int_A \int g d\Pi_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{pushforward}) \\ &= \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration}) \\ &= \int_S g(F) dP. \end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. $\square$

Corollary 2.4 (Predictive sufficiency). $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e.

Proof. $\sigma(Q_*) \subseteq \sigma(Q)$, and the conditional expectation given $\sigma(Q)$ is already $\sigma(Q_*)$ -measurable by the theorem. $\square$

The operator $(K_{Q_*}g)(q_*) = \int g dq_*$ on $\mathcal{P}(\beta)$ recasts the theorem as: $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e.

3 Dynamics and duality

To introduce dynamics, index over a discrete time horizon. For each $t \in \mathbb{N}$, let $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ be a Markov kernel on the state space γ , with $\Pi_0(q, \cdot) = \delta_q$. A family $\{\Pi_t\}$ satisfying the Chapman–Kolmogorov equation

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s)$$

is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \text{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 3.1 (Semigroup property). $K_{t+s}g = K_t(K_s g)$ for all bounded measurable g .

Proof. $(K_{t+s}g)(q) = \int g d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \int (\int g d\Pi_t(q', \cdot)) d\Pi_s(q, q') = \int (K_t g)(q') d\Pi_s(q, q') = (K_t(K_s g))(q)$, where the exchange of integrals is justified by Bochner–Fubini. $\square$

The pushforward of a measure μ on γ through the kernel is $P_t^* \mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot) d\mu(q)$.

Theorem 3.2 (Koopman–Perron duality). For every bounded measurable g and finite measure μ :

$$\int (K_t g) d\mu = \int g d(P_t^* \mu).$$

Proof. $\int g d(P_t^* \mu) = \int g d(\mu \star \Pi_t) = \int_\gamma \int g d\Pi_t(q, \cdot) d\mu(q) = \int (K_t g) d\mu$, by Bochner–Fubini. $\square$

When $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$, the stochastic picture collapses.

Proposition 3.3 (Deterministic limit). If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$, then $K_t g = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.

Proof. $(K_t g)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. $\square$

In the deterministic case, $K_t g = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_t g d\mu = \int g d(P_t^* \mu)$ becomes the change-of-variables formula $\int g \circ T^t d\mu = \int g d(T_*^t \mu)$.

The full structure produced by the prediction construction is captured in one definition.

Definition 3.4 (Observable dynamical system). An *observable dynamical system* is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ where $\mathcal{P}(\beta)$ is the predictive state space (the image of Q_*), $\nu_{Q_*} = P \circ Q_*^{-1}$ is the pushforward of P along Q_* , and $\{K_t\}$ is the Markov semigroup on $\mathcal{P}(\beta)$.

4 From prediction to reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n \omega))_{n \in \mathbb{Z}}$ is the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The predictive state space $\Phi_h(X)$ is a subset of $\mathbb{R}^\mathbb{Z}$ encoding the dynamical information that h can see. Whether it is a faithful image of X depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X . Injectivity of Φ_h modulo μ -null sets is equivalent to the observable σ -algebra

$$\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$$

equalling $\mathcal{B}(X)$ modulo μ .

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable σ -algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$. The next two sections answer this question.

5 Density bridge and reconstruction

Throughout this section and the next, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \rightarrow X$ is a measurable, μ -preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

Definition 5.1 (Observable algebra). The *observable algebra* generated by h is

$$\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) \subseteq \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 5.2 (Delay map). The *delay map* is

$$\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}, \quad \Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n , the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that Φ_h is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞ .

Lemma 5.3 (Density bridge). *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ .

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$. Every $f \in \mathcal{A}$ is $\mathcal{A}$ -measurable, hence $\mathcal{M}$ -measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$. It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V . By the functional monotone class theorem [1], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \rightarrow f$ pointwise a.e. with $f \in L^\infty$, then $f_n \rightarrow f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$ -indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$ -simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B} \bmod \mu$. $\square$

Remark 5.4 (Role of the algebra structure). The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed *linear* span $\mathcal{H}_\mathcal{A} = \overline{\text{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_\mathcal{A} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu = \text{Lebesgue}$, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ -algebra (the dyadic structure separates points). But $\text{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2 . So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B} \bmod \mu$ — see Corollary 6.2.

Theorem 5.5 (Reconstruction theorem). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

- (i) $\mathcal{O}_h = \mathcal{B}$ modulo μ ;
- (ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;
- (iii) $\Phi_h : X \rightarrow \mathbb{R}^\mathbb{Z}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$ modulo μ .

Proof. (i) $\Leftrightarrow$ (ii). Apply the density bridge (Lemma 5.3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

(i) $\Leftrightarrow$ (iii). By Definition 5.2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z}))$. So $\mathcal{O}_h = \mathcal{B} \bmod \mu$ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B} \bmod \mu$, which is exactly condition (iii). $\square$

Remark 5.6 (What “measure-theoretic embedding” means). Condition (iii) says that Φ_h is *almost everywhere injective* and the pullback σ -algebra is all of $\mathcal{B}$. More precisely: since $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B} \bmod \mu$, for any two sets $A, A' \in \mathcal{B}$ with $\mu(A \triangle A') > 0$ there exists a Borel set $E \subseteq \mathbb{R}^\mathbb{Z}$ with $\mu(\Phi_h^{-1}(E) \triangle A) = 0$ and $\mu(\Phi_h^{-1}(E) \triangle A') > 0$. In measure-theoretic language, Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^\mathbb{Z}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^\mathbb{Z})/(\Phi_h)_*\mu)$.

The T -invariance of μ implies that $(\Phi_h)_*\mu$ is invariant under the bilateral shift σ on $\mathbb{R}^\mathbb{Z}$, since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$ for each n . The dynamical system (X, T, μ) is therefore isomorphic to $(\mathbb{R}^\mathbb{Z}, \sigma, (\Phi_h)_*\mu)$ restricted to the image of Φ_h .

6 Cyclic vectors and Stone space identification

Definition 6.1 (Cyclic vector). The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h := \overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu).$$

Corollary 6.2 (Cyclic implies reconstruction). *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T , then $\mathcal{O}_h = \mathcal{B}$ modulo μ .*

Proof. Since $\mathcal{A}_h$ contains $\text{span}\{h \circ T^n\} = \text{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. $\square$

Remark 6.3 (The implication is strict). The converse of Corollary 6.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h , the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2 -spectrum*: $L^2(X, \mu)$ is a cyclic U_T -module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [2] but is not universal.

The Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \rightarrow \text{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Proposition 6.4 (Stone space identification). *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ . Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\text{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\text{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. Injectivity mod μ . Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\supset x'$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \triangle F) \cup F^c$, placing x' in a null set. Hence ι is injective μ -a.e.

σ -algebra identification. The pullback ι^{-1} maps the Baire σ -algebra of $\text{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

Measure recovery. Since μ is σ -additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [4] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. $\square$

The Stone space introduced in [4] as a compact ambient space for deriving σ -additivity is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

The classical Takens embedding theorem [6] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d -manifold and a generic smooth h , the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \dots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d + 1$. The reconstruction theorem here requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h , the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2 -dense: the density condition holds and the two results are compatible.

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